- **Loss function** $\mathrm{L}(y, \hat{y})$:
  Measures error for a **single training example**.
  Example (squared loss):

$$\mathrm{L}(y^{(i)}, \hat{y}^{(i)}) = (y^{(i)} - \hat{y}^{(i)})^2$$

- **Cost function** $J(\theta)$:
  Measures the **average error across the whole dataset** (all training examples).
  It is the objective we try to minimize during training.

🧮 Relationship: Cost Function = Mean of Losses

If we have:

- Dataset with m examples: $\{(x^{(i)}, y^{(i)})\}_{i=1}^{m}$
- Hypothesis $h_\theta(x)$ predicting $\hat{y}^{(i)}$

Then:

$$J(\theta) = \frac{1}{m} \sum_{i=1}^{m} \mathrm{L}(y^{(i)}, h_\theta(x^{(i)}))$$

> ✅ So, **cost function is just the average loss over the dataset**.

📌 Examples

1. **Linear Regression**

- Loss (squared error):

$$\mathrm{L}(y^{(i)}, \hat{y}^{(i)}) = (y^{(i)} - \hat{y}^{(i)})^2$$

- Cost:

$$J(\theta) = \frac{1}{2m} \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)})^2$$

> The $\frac{1}{2}$ is added for mathematical convenience during differentiation.